
routing
A* on the lane graph, on request
reference line
lane centre 180 to 250 m ahead, smoothed: QP every 50 ms
path QP
piecewise jerk, a station every 0.5 m over 100 m
speed: DP, then QP
search, then piecewise jerk, a point every 0.1 s over 7 s
trajectory
8 s, once per prediction message
parking (open space): Hybrid A* path → smoothing QP → speed QP, once per manoeuvre